Chaotic Time-Series Forecasting: A Comparative Study of Attention-Based Models and LSTM

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Table of Contents

Abstract 3

1 Research Objective and Background 3

2 Systems, Data and Forecasting Tasks 3

3 Development of the Forecasting Pipeline 4

4 Forecasting Models and Training Objectives 8

5 Evaluation of Autonomous Forecasts 9

6 Results and Analysis 11

7 Discussion 15

8 Conclusions and Future Work 16

References 16

Appendix A Verified Experimental Parameters 17

Appendix B Earlier Exploratory KS Runs 19

Abstract

This report studies forecasting at observed spatial positions in the chaotic Kuramoto–Sivashinsky (KS) system. Its central comparison uses three recorded forecasting procedures—self-attention, Easy Attention with a multiplicative feature gate, and LSTM—when spatial sampling leaves four channels (ds = 16) or two (ds = 32) from a 64-point grid. I first explain how the project changed its data split, attention rollout, error normalization and recurrent training. I then recalculate forecast duration from the unsmoothed error curve of each of 20 held-out validation trajectories. At the more permissive NRMSE threshold of 1, mean valid prediction time is 1.465, 1.285 and 1.098 Lyapunov times for the feature-gated Easy Attention, self-attention and LSTM procedures at ds = 16. At ds = 32, self-attention leads at 1.227, followed by feature-gated Easy Attention at 1.083 and LSTM at 0.681. The stricter threshold of 0.5 gives a different ordering of the two attention procedures at ds = 32. These first-crossing times describe brief initial agreement; the ds = 32 LSTM loses trajectory accuracy early. Earlier dense-observation experiments used a different split and NRMSE denominator and serve as background. Independent Lorenz–96 and two-dimensional Kolmogorov tests extend the study to separate tasks and time scales. The report traces both the revised methods and the measured limits of long autonomous forecasts.

1 Research Objective and Background

A deterministic chaotic system can amplify small differences in its initial state. A neural forecaster may follow one particular trajectory for a limited interval and then lose its phase, even while producing values that still look plausible. This study asks how its error grows after true future states are withheld, and how that growth changes when spatial observations become sparse.

The KS task uses historical values at measured positions to forecast future values at those same positions. At ds = 32, the forecast contains two observed channels. I compare three complete forecasting procedures, including their encoders, training objectives, warm-up lengths and rollout rules. The report also follows the methodological changes that make their results interpretable.

Self-attention learns data-dependent interactions among positions in an observed sequence [3]. Easy Attention uses learned temporal weights [4]; the feature gate in this project is described from its implementation. Recurrent and reservoir methods have been used for complex spatiotemporal forecasts, with error-based forecast horizons [2]. Learning from a model's predictions provides background for closing the training–inference gap [8]. The LSTM here uses its own staged closed-loop schedule.

2 Systems, Data and Forecasting Tasks

2.1 Kuramoto–Sivashinsky observations

The numerical reference follows the one-dimensional KS equation on a periodic interval:

$$u_t + uu_x + u_{xx} + u_{xxxx} = 0, u(x + L, t) = u(x, t)$$

Here u(x,t) is a scalar state, subscripts denote partial derivatives, and the configured periodic domain length is L = 22. The associated generator uses a 64-point Fourier pseudospectral ETDRK4 solver [1] with a 0.1 model-time step and records states at the same 0.1 interval. It creates randomized initial arrays and applies a numerical warm-up. The quantitative results below are recalculated from recorded predictions, true forecast windows and run metrics.

Let N = 64 be the full spatial grid size and ds be the integer grid-point stride. The grid spacing and number of observed channels are

$$\Delta x = \frac{L}{N}, D = \frac{N}{\mathrm{ds}}.$$

At each saved time t, the network receives the observation vector

$$\mathbf{x}_t = [u(0,t), u(\mathrm{ds}\Delta x, t), \ldots, u((D-1)\mathrm{ds}\Delta x, t)] \in \mathbb{R}^D.$$

The stride ds counts grid points between observations; the corresponding physical spacing is dsΔx. Thus ds = 16 gives D = 4 and ds = 32 gives D = 2; ds = 1 gives all 64 observed channels. Spatial sampling leaves the saving interval unchanged. One trajectory is a time-ordered realization of these observed states. A training window is a shorter history and target cut from a trajectory, and windows within one realization can overlap.

2.2 Scope of the supplementary systems

For Lorenz–96, x_j is the j-th component of a cyclic 40-variable state, F = 8 is constant forcing, and indices wrap around the ring:

$$\frac{dx_j}{dt} = (x_{j+1} - x_{j-2})x_{j-1} - x_j + F$$

A fourth-order Runge–Kutta integrator saves states every 0.01 model-time units. The supplementary models forecast the full observed 40-component state on 20 recorded independent test trajectories for 3,000 saved steps. The configured λ = 1.68 comes from published work on this forcing regime [2].

The two-dimensional Kolmogorov experiment instead predicts a vorticity field ω(x,y,t) on a 64 × 64 periodic square $[0,2\pi)^2$. Its generator implements

$$\omega_t + u\omega_x + v\omega_y = \nu\Delta\omega - \alpha\omega - n_f\cos(n_f y), u = \psi_y, v = -\psi_x, \omega = -\Delta\psi$$

The configured viscosity ν = 0.005, linear drag α = 0.1 and forcing wavenumber n_f = 4 define this run. A pseudospectral RK4 scheme uses internal step 0.005 and saves every 0.01. These field models add a convolutional spatial encoder and decoder around the temporal forecaster. Evaluation records identify 15 separate, 2,000-step test trajectories. Their Lyapunov exponent setting is uncalibrated, so the results below use saved steps and model time.

3 Development of the Forecasting Pipeline

The revised experiments changed four connected parts of the workflow: the data split, attention rollout, error scale and recurrent training. This section traces those changes through the initial and final implementations.

3.1 Split trajectories before constructing windows

In the initial attention pipeline, the code first cut overlapping windows from all 100 recorded trajectories and then randomly assigned windows to training or validation. Its input length was 64 saved steps and its stride was 8, so adjacent input windows shared 56 states. Windows from the same trajectory could land in both groups. The resulting validation score partly reflected interpolation within trajectories already present in training and could overstate skill on a new trajectory.

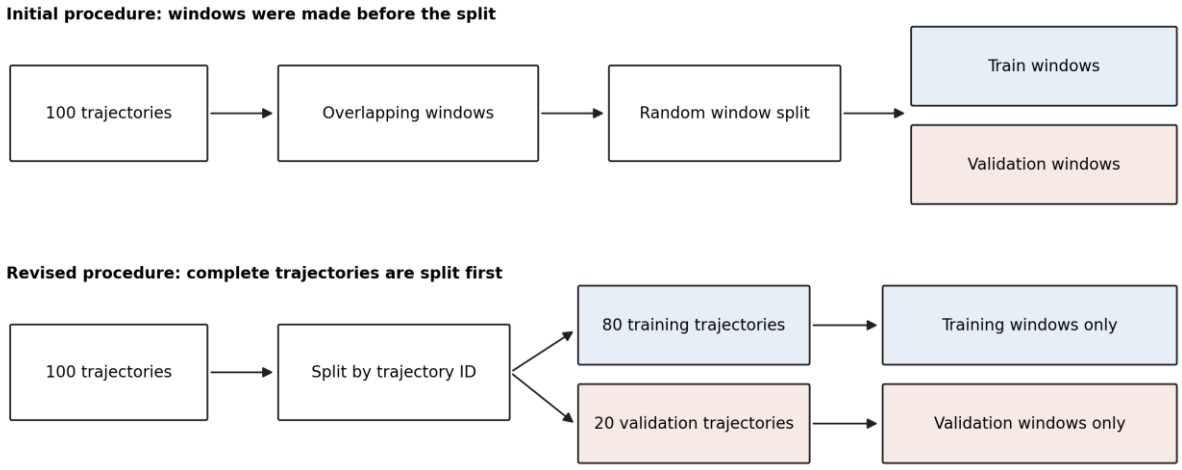


Figure 1. Order of operations in the initial and revised KS splits. The revised procedure assigns complete trajectory IDs before windows are made; colors distinguish the resulting sets.

The revised loader assigns 80 complete trajectory IDs to training and 20 to validation, then extracts windows within each group. Its per-channel normalization is fitted on the 80 training trajectories. Each trajectory now belongs to one group, removing the overlap of its windows across training and validation. The 20 held-out trajectories serve in model selection, so the KS results below are validation results. The earlier exploratory curves came from the window-level split and remain useful for describing the project's development.

3.2 Use the attention model's whole predicted block

The initial attention model took 64 observed time steps and was supervised to output three future steps. During inference, the script kept the first output, inserted that prediction into the next history and called the model again. The second and third supervised outputs were discarded in the long forecast. Its farthest directly supervised target was step 3. A 1,200-step rollout required about 1,200 model calls, one per predicted step.

The revised attention runs take 128 observed steps, directly predict the next 100 steps, and feed all 100 into the following input window. The next input joins the old history with the predicted block and keeps the most recent 128 states. The farthest directly supervised target is step 100. Each block is produced in one call; later blocks depend on earlier predicted blocks, making the long forecast autoregressive. A 1,200-step rollout needs 12 block calls. This is a call-count comparison; elapsed runtime and accuracy depend on the larger block and other changed settings.

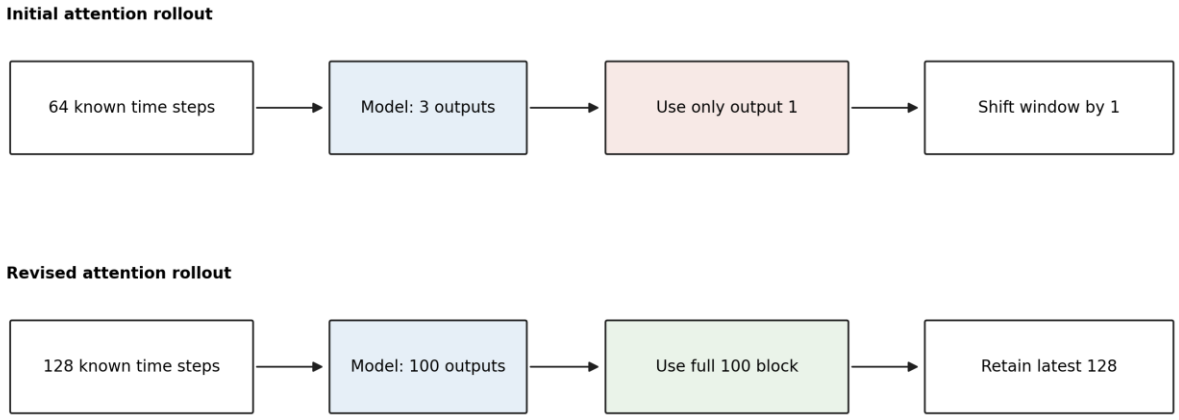


Figure 2. Attention-model rollout before and after the revision. The numbers 64, 3, 128 and 100 are temporal lengths; the spatial channel count D is separate.

Stage	Initial attention procedure	Revised attention procedure
Input history	64 saved steps	128 saved steps
Supervised output	3 future steps	100 future steps
Used at inference	Only output 1	All 100 outputs
Window advance	1 saved step	100 saved steps
Calls for 1,200 steps	About 1,200	12

Table 1. Temporal rollout operations in the initial and revised attention implementations.

3.3 Replace a changing spatial denominator with training-channel scales

The initial NRMSE compared each predicted snapshot with its true snapshot, but normalized by the spatial variance of that particular true snapshot. For validation trajectory i, forecast step k, D observed channels j, and stabilizer $\varepsilon = 10^{-8}$, the old calculation was

$$\mathrm{MSE}_{i,k} = \frac{1}{D}\sum_{j=1}^{D}\left(\hat{u}_{i,k,j} - u_{i,k,j}\right)^2$$

$$\mathrm{Var}_{\mathrm{space},i,k} = \frac{1}{D}\sum_{j=1}^{D}\left(u_{i,k,j} - \overline{u}\right)^2, E_{i,k}^{\mathrm{old}} = \sqrt{\frac{\mathrm{MSE}_{i,k}}{\mathrm{Var}_{\mathrm{space},i,k} + \varepsilon}}$$

Here ū is the mean across the D true spatial channels at that forecast step. The old denominator is the spatial variance of the current true snapshot plus ε. At ds = 32, D = 2: when the two observed true values approach each other, that variance approaches zero and the score can rise sharply even for modest absolute prediction errors. The denominator also changes at every forecast step, making a curve's height depend on the evolving spatial spread of the truth.

The revised procedure calculates the mean μ_j and population standard deviation σ_j for each retained channel j over only the training trajectories and their time samples. It uses these statistics to standardize model inputs. For evaluation, prediction and truth are compared in physical value units, their signed difference is divided once by the same fixed σ_j, and the channel-normalized errors are combined:

$$E_i(k) = \sqrt{\frac{1}{D}\sum_{j=1}^{D}\left(\frac{\hat{u}_{i,k,j} - u_{i,k,j}}{\sigma_j}\right)^2}, \overline{E}(k) = \frac{1}{M}\sum_{i=1}^{M} E_i(k)$$

The code rejects a channel with near-zero training standard deviation and uses $\max(\sigma_j, 10^{-8})$ for numerical protection. The resulting score measures error against each channel's typical variation in training and keeps that scale fixed across forecast leads. Changing the denominator changes the score, so the old and revised NRMSE curves require separate interpretation.

Spatial sampling changes D. The mean curve $\bar{E}(k)$ averages each trajectory's already normalized root mean square error, preserving the calculation order used in the VPT analysis.

3.4 Train the LSTM on its own predictions

The final LSTM training warms the hidden and cell states on known history and then feeds each predicted state back for H steps, matching the feedback mechanism used during autonomous forecasting. The training horizon grows in stages from 10 to 20, 50 and 100 steps. Gradients propagate through each closed-loop segment, exposing the model to errors from its earlier predictions. This local schedule addresses the training–inference mismatch discussed in [8].

Epochs	Closed-loop horizon H	Learning rate at stage start
1–20	10 steps	5×10^{-4}
21–50	20 steps	3×10^{-4}
51–100	50 steps	2×10^{-4}
101–250 (planned)	100 steps	1×10^{-4}, cosine toward 2×10^{-5}

Table 2. LSTM training schedule in the final run configuration. The last row is the planned maximum; actual logs and selected checkpoint epochs are recorded separately in Appendix A.

The current-H training loss changes its prediction target when H changes. A rise at a stage boundary reflects the longer rollout now being optimized. The saved ds = 16 and ds = 32 training traces make those transitions visible. Section 6 also follows a fixed-100-step validation diagnostic through training; the separate 3,000-step NRMSE curves assess long-rollout skill.

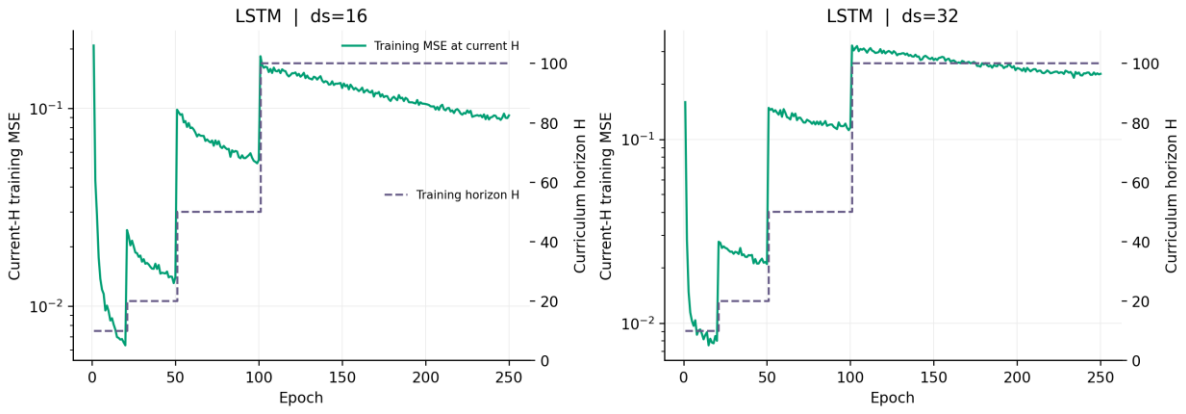


Figure 3. LSTM current-H training loss and configured horizon for ds = 16 and 32. H changes after epochs 20, 50 and 100; loss values from different stages measure different rollout lengths.

4 Forecasting Models and Training Objectives

I compare the three final model implementations. Each predicts the D spatial positions supplied as observations. The attention models process an observed history as a sequence and emit a 100-step block; the LSTM carries recurrent memory through a one-step forecast. These operations determine how an early error enters later inputs, so I describe the implemented calculations before comparing their curves.

4.1 Multi-head self-attention

Each raw training input X contains 128 observed time positions and D channels, where D is four at ds = 16 and two at ds = 32. A learned input projection maps each position to 256 features and adds a sinusoidal position signal. In the following layer equations, the raw input and the feature matrix entering an encoder layer have shapes

$$X \in \mathbb{R}^{128\times D}, \tilde{X} \in \mathbb{R}^{128\times 256}.$$

The first layer receives the projected input; later layers receive the preceding layer's output. Three encoder layers are used. In each layer, eight heads divide the 256 features into head width $d_h = 32$. For head h, the query, key and value matrices are

$$Q_h = \tilde{X}W_{Q,h}, K_h = \tilde{X}W_{K,h}, V_h = \tilde{X}W_{V,h}.$$

Each conceptual per-head weight block has shape 256 × 32, giving 128 × 32 queries, keys and values. The implementation uses biased 256-to-256 linear maps and then reshapes their outputs into eight heads; the displayed projections omit those biases.

$$S_h = \text{softmax}(\frac{Q_h K_h^T}{\sqrt{d_h}}), Z_h = S_h V_h, d_h = 32$$

For each head, the weight matrix S_h has shape 128 × 128; softmax is taken across its last dimension, so one row weights the observed history for one observed position. The weighted output Z_h has 128 rows of 32 features. The weights respond to the particular input because both Q_h and K_h do. The eight outputs are joined, projected back to width 256, and passed through a residual connection, layer normalization and a GELU feed-forward network of width 1,024. Every position in the observed history is available to the encoder, so it uses no causal mask. The 100 future target states are withheld from Q, K and V.

After the third layer, the encoded feature vector at the last observed position enters a small prediction head. The head produces 100 × D numbers and reshapes them into the next 100 observed states in one call. Autonomous prediction starts when the whole block is appended to the history and the newest 128 positions are supplied for the next call. Thus the model uses real history for the first block and increasingly uses its own outputs at later block boundaries.

4.2 Easy Attention with a feature gate

The second attention model retains the 256-feature input projection, positional signal, three encoder layers, eight heads and 100-step output head. Within each layer and head it learns a matrix α_h of size 128 × 128, initialized near the identity. This matrix mixes projected values across the 128 observed positions. The layer uses a shared, unconstrained α_h in place of query and key projections and softmax scores. With head width 32 and the layer feature matrix defined above, the essential operations are

$$V_h = \tilde{X}W_{V,h}, Z_h = \alpha_h V_h.$$

$$G = \text{GELU}(\tilde{X}W_U).$$

$$Y = (\text{Concat}(Z_1, \ldots, Z_8) \odot G)W_O$$

Here ⊙ is elementwise multiplication. These are schematic equations; the implemented projections include biases. The shared α_h mixes time positions, while V_h and G vary with the input. The gate projection maps 256 features to 256 at each time position, so G has shape 128 × 256 and matches the concatenated values. It multiplies those values before an output projection, residual connection, normalization and feed-forward network. The α_h entries are unconstrained real values, so they should be read as learned mixing weights.

The implemented gate acts across features at each time position; the published gMLP Spatial Gating Unit [5] includes a cross-position projection in the gating branch. I call the implemented operation a "feature gate." The model combines this multiplicative gate with learned Easy Attention temporal scores [4]. The final comparison has no matched Easy Attention run with the gate removed, so its individual contribution remains unmeasured.

4.3 LSTM

The recurrent comparison uses three stacked LSTM layers [6] with 256 hidden units per layer and a linear projection back to D observed channels. Its hidden state h and cell state c carry information forward. In one LSTM cell, the forget gate f chooses how much old cell content remains, the input gate i admits a candidate update g, and the output gate o controls the exposed hidden state. Their state updates are

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t, h_t = o_t \odot \tanh(c_t)$$

The first call processes a true observed warm-up of 128 steps at ds = 16 or 256 at ds = 32. Its last output is the first future prediction. Every later call takes the previous predicted state as input and carries forward the hidden and cell states. The plotted 3,000-step result is an autonomous one-step rollout after the initial warm-up. The longer warm-up at ds = 32 is a relevant difference between these runs.

The code runs this LSTM in float32, with no residual links between recurrent layers. During training it drops an input feature with probability 0.005 and, when returning hidden and cell states from a call, retains an old state element with probability 0.02. This local form of zoneout [7] uses a keep-probability setting of 0.98 for taking the new state. Both stochastic operations are disabled during the plotted inference rollout. The multi-step warm-up is one PyTorch LSTM call; the local zoneout mask applies to its returned terminal state.

The attention runs compare one directly produced 100-step block with all 100 target positions. The LSTM is warmed on true history, then trained through the predictions it feeds back to itself. Its closed-loop horizon H increases from 10 to 20, 50 and 100 steps at the stages in Table 2. Figure 3 shows the current-H training loss and the training horizon for ds = 16 and 32; Figure 8 shows the fixed-100-step validation loss. Each horizon change makes the training task longer, so a jump at a stage boundary needs to be read against the new H. The 3,000-step rollout remains a separate evaluation of long-term behavior beyond the final 100-step training horizon.

5 Evaluation of Autonomous Forecasts

5.1 Error at each forecast step

I evaluate a prediction against the true future at the same observed positions, after returning both to physical units. The fixed scale σ_j is the population standard deviation of observed channel j calculated from the training trajectories, as explained in Section 3.3. For validation trajectory i and one-based forecast step k, the per-trajectory error is

$$E_i(k) = \sqrt{\frac{1}{D}\sum_{j=1}^{D}\left(\frac{\hat{u}_{i,k,j} - u_{i,k,j}}{\sigma_j}\right)^2}$$

The KS comparison plots display the mean of $E_i(k)$ over the same 20 held-out validation trajectories at each lead time. They use the revised fixed training-channel denominator. The comparison figures retain a forecast-step axis to show the shape and late behavior; the forecast-duration summary uses Lyapunov units. These trajectories sample forecast starts from one fitted run per model.

5.2 First-crossing valid prediction time

An averaged curve can conceal differences in the useful interval of individual trajectories. I calculate a first crossing for every unsmoothed $E_i(k)$. For threshold a equal to 0.5 or 1, the crossing step $k^*_{i,a}$ is the first forecast step where that trajectory's error reaches or exceeds a. This definition fixes the valid interval at its first crossing, even if the error later falls. In all six principal KS groups, each of the 20 trajectories crosses both thresholds before step 3,000, so the reported means require no censoring rule.

$$k^*_{i,a} = \min\{k \geq 1 : E_i(k) \geq a\}$$

$$\mathrm{VPT}^{\lambda}_{a} = \lambda_{\mathrm{ref}}\Delta t\frac{1}{M}\sum_{i=1}^{M} k^*_{i,a}\,, a \in \{0.5,1\}$$

Here M = 20 for the KS validation set. The chosen KS reference exponent is $\lambda_{ref} = 0.043$ for a periodic length-22 system [9], and the saved interval is $\Delta t = 0.1$; each forecast step therefore equals 0.0043 reference Lyapunov times. The exponent is a literature and configuration value, with no new fit to the present trajectories. VPT averages the individual crossing steps before converting units. For the ds = 32 LSTM, the mean individual first crossing of NRMSE 1 is 158.3 steps, or 0.681 Lyapunov times; the mean curve first crosses 1 at step 418. Its finite VPT measures a brief early interval of acceptable error, while its long autonomous rollout fails to track the true trajectories.

I also checked how the ranking responds to a more persistent crossing rule: the error had to remain above the threshold for 10 consecutive forecast steps. This diagnostic preserved every model ranking within each ds and threshold. It did change the size of some gaps. At ds = 32 and threshold 0.5, the Easy Attention–self-attention difference fell from 7.55 to 2.20 steps. All reported VPT values continue to use the unsmoothed first crossing defined above.

5.3 Reading the 1 and √2 guide lines

Figures 4 and 6 show horizontal guides at 0.5 and 1 for the two VPT thresholds, while √2 serves as a long-lead reference. If standardized prediction and truth have approximately unit variance, similar means and negligible correlation, the expected squared normalized difference for one channel is about 2, giving a root scale near √2. Here the plotted mean is formed from only 20 per-trajectory square roots over two or four observed channels, so it has no exact √2 expectation.

Late values around √2 can change with forecast amplitude, bias and correlation, and fluctuations across that line do not establish convergence. The ds = 32 LSTM has a mean curve near NRMSE 1 later in the rollout. Its individual first crossings show a short initial accurate period, followed by failure to track the true trajectories over long leads.

5.4 Validation, checkpoint choice and comparison limits

For KS, the trajectory split assigns 80 full realizations to training and 20 to validation before windows are made. These 20 validation trajectories appear in the NRMSE plots and contribute to model selection. In the 1,200-step validation rollout, a predicted-to-true temporal standard-deviation ratio below 0.5 over the last 200 steps flags amplitude collapse; for LSTM, a spectral-entropy ratio below 0.65 over the last 1,000 steps also flags repetitive output. Checkpoint ranking then considers the mean error curve's crossings, its integrated error and validation loss. The selected 3,000-step KS plots describe held-out validation trajectories. Only one training run is reported for each model and sampling factor, so variability across training seeds was not assessed.

Within each KS sampling factor, the runs share observed positions, the training-channel NRMSE rule, the 20 validation trajectory identities and a 3,000-step evaluation horizon. The ds = 32 LSTM uses a longer true warm-up; attention feeds back 100-step blocks and LSTM feeds back one step at a time. The training objectives and checkpoint epochs also differ. VPT comparisons therefore assess each complete procedure. The separate contributions of recurrence, α and the feature gate remain unresolved.

The Lorenz–96 and Kolmogorov extensions use separately generated independent test trajectories. Their equations, observation dimensions and prediction pipelines differ from KS, so I discuss their results within each system. Lorenz–96 uses a published F = 8 reference exponent of about 1.68 [2], which was not measured on these trajectories. The Kolmogorov configuration sets λ = 1.0 and marks it uncalibrated; I report its first crossings in forecast steps or saved model-time units. Section 6 presents these as separate system checks.

6 Results and Analysis

6.1 KS: short-horizon skill and first-crossing times

The six principal runs contain one selected checkpoint for each model and each of the two sparse observation settings. Every curve is evaluated on 20 held-out validation trajectories for 3,000 saved forecast steps. Figure 4 expands the first 600 steps, where the differences relevant to VPT appear; Figure 6 shows the full horizon. The horizontal guides at 0.5 and 1 mark the two chosen VPT tolerances, while √2 helps interpret late error. The plotted curves average errors at each lead, and Table 3 averages the individual trajectories' first crossings.

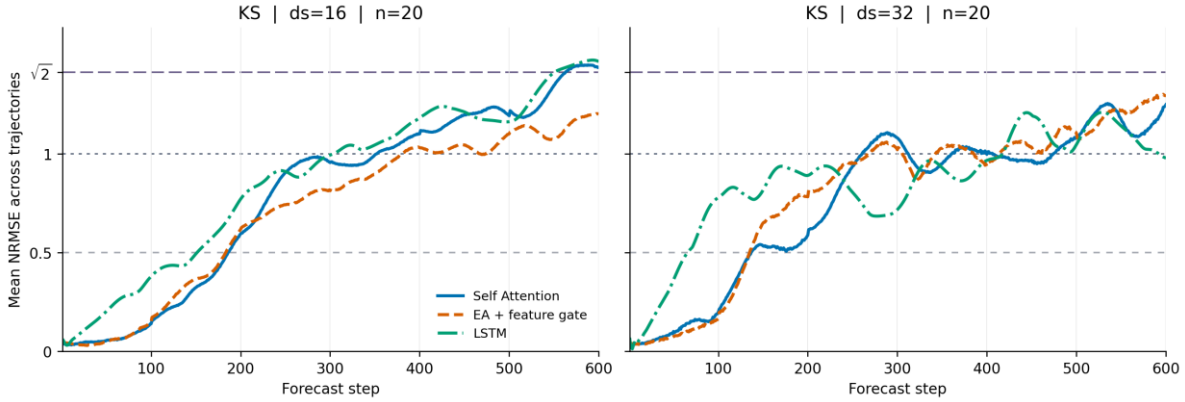


Figure 4. Early KS forecast error. Each panel compares three procedures at one sampling factor; curves are mean NRMSE across 20 held-out validation trajectories at each saved forecast step (Δt = 0.1). Guides mark 0.5, 1 and √2. The first 600 steps are displayed without changing the underlying error data.

ds / channels	Forecasting procedure	$VPT_{0.5}$ (LT)	VPT_1 (LT)
16 / 4	Self-attention	0.9090	1.2846
16 / 4	Easy Attention + feature gate	0.9746	1.4652
16 / 4	LSTM	0.6699	1.0978
32 / 2	Self-attention	0.7471	1.2266
32 / 2	Easy Attention + feature gate	0.7796	1.0825
32 / 2	LSTM	0.4358	0.6807

Table 3. Mean of per-trajectory, unsmoothed first NRMSE crossings, converted to KS reference Lyapunov times (LT) using 0.043 × 0.1 per saved step. The chosen tolerances are 0.5 (stricter) and 1 (more permissive). All 20 trajectories cross both thresholds in each run.

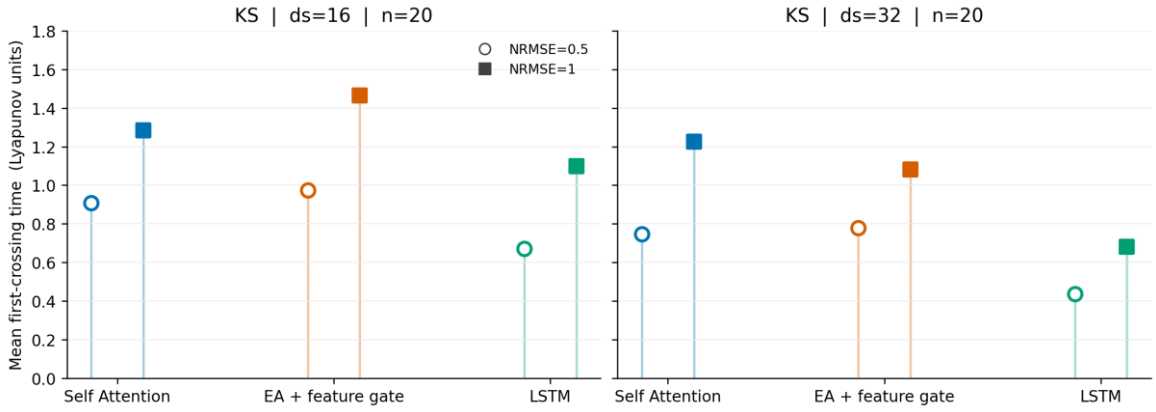


Figure 5. KS valid prediction times from Table 3. Markers show means of the 20 individual first crossings at NRMSE 0.5 and 1, expressed in reference Lyapunov times.

At ds = 16, the feature-gated Easy Attention procedure has the longest mean time at both tolerances. Its VPT_1 of 1.4652 LT exceeds self-attention's 1.2846 LT by 0.1806 LT and LSTM's 1.0978 LT by 0.3674 LT. At the stricter 0.5 tolerance, the corresponding means are 0.9746, 0.9090 and 0.6699 LT. These differences describe the complete recorded procedures and selected checkpoints.

At ds = 32, the two attention procedures exchange order when the tolerance changes. At NRMSE 0.5, feature-gated Easy Attention reaches 0.7796 LT compared with self-attention's 0.7471 LT. At NRMSE 1, self-attention reaches 1.2266 LT, ahead of feature-gated Easy Attention at 1.0825 LT. The LSTM values, 0.4358 and 0.6807 LT, are shorter under both choices. The reversed attention ordering shows why the threshold must be stated beside every VPT claim. The ds = 32 LSTM uses 256 warm-up steps, while the attention models use 128, so most of its truth windows start at a later point within the same held-out trajectory; this limits a strictly paired interpretation of the ds = 32 ranking.

6.2 Long rollouts and representative state values

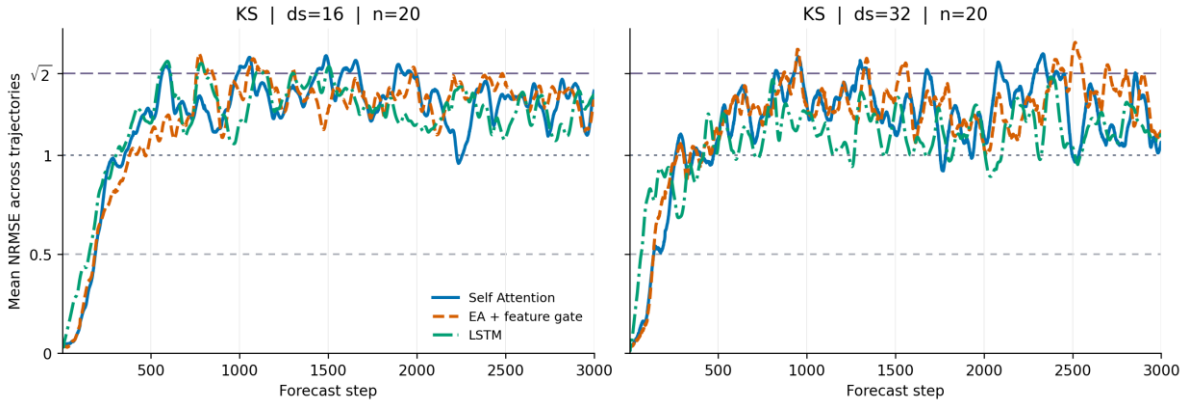


Figure 6. KS mean NRMSE over all 3,000 saved forecast steps, averaged across 20 validation trajectories, with 0.5, 1 and √2 guides. The attention curves oscillate around √2 at late leads.

All six mean error curves rise sharply and then vary substantially over the long rollout. The initial trajectory agreement disappears well before step 3,000. The attention curves repeatedly move around √2; their continuing fluctuations and finite-trajectory average give no clear convergence to that value. Their late behavior is more accurately described as persistent error near the scale of decorrelated states.

The ds = 32 LSTM begins with a usable prediction interval: every evaluation trajectory starts below NRMSE 0.5 and later crosses both thresholds. Individual NRMSE reaches 1 after a mean 158.30 saved steps (0.6807 LT). The mean curve first reaches 1 at step 418, showing that its crossing overstates the interval captured by per-trajectory VPT. The mean curve stays below √2 throughout the 3,000-step horizon, with a maximum of 1.3956 and a last-500-step mean of 1.1287. Useful pointwise tracking is lost much earlier than these late mean values suggest.

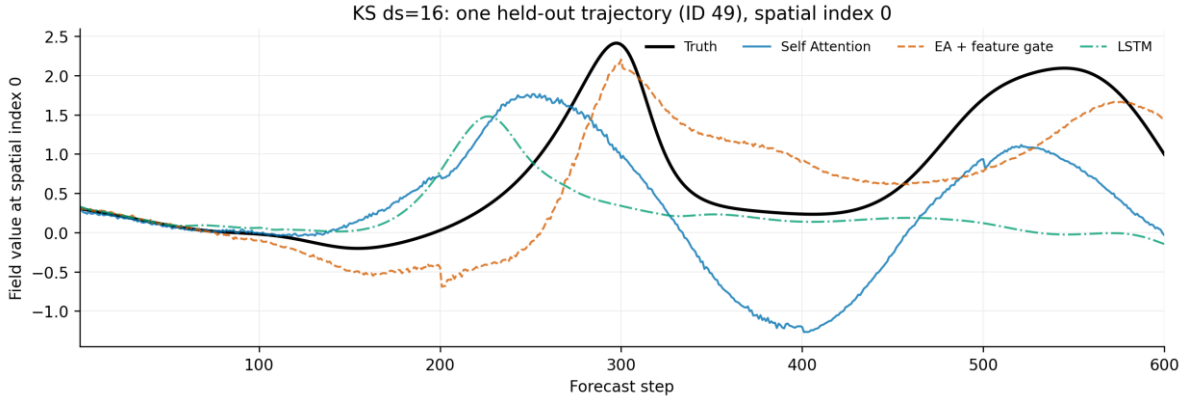


Figure 7. First listed held-out KS trajectory (ID 49), observed channel 0 at ds = 16, over the first 600 saved forecast steps. The three procedures share the same true forecast window. Initial agreement gives way to phase loss.

Figure 7 shows what the error growth means at one observed position. Each model initially follows the true state to some extent, then its peaks and troughs lose alignment as feedback continues. Oscillations can retain a plausible amplitude after losing the correct phase. The trace complements the 20-trajectory NRMSE and VPT summaries.

6.3 What the training curves establish

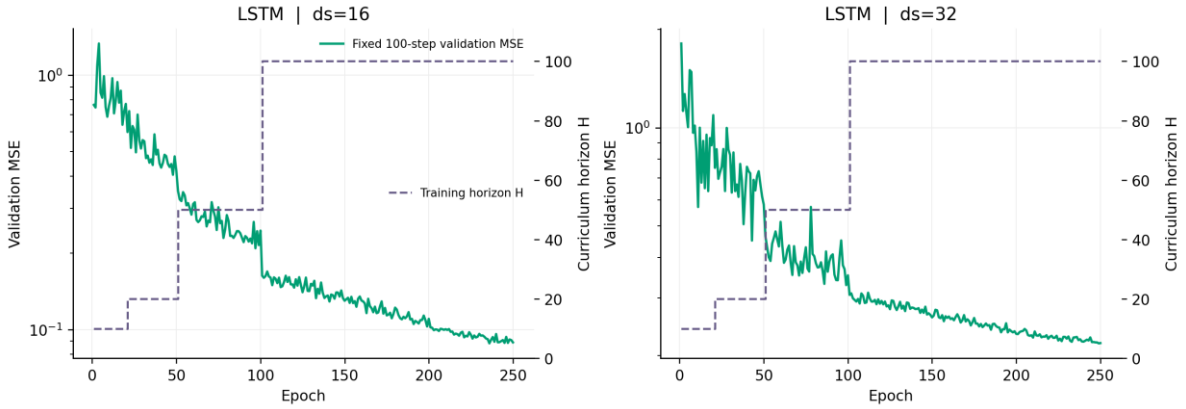


Figure 8. LSTM fixed-100-step closed-loop validation MSE and training horizon H, shown for ds = 16 and 32. The validation target stays at 100 steps while H increases after epochs 20, 50 and 100.

The current-H loss in Figure 3 measures the task presented to LSTM at each epoch; the fixed-100-step validation curve in Figure 8 follows one target through training. A rise in current-H loss at a stage boundary accompanies the longer feedback horizon. Attention validation loss covers a directly predicted 100-step block, while LSTM validation loss covers a closed-loop segment, so the two losses address different training objectives. The saved checkpoints were selected using held-out rollout behavior over a configured 1,200-step horizon. LSTM selected epoch 245 at ds = 16 and 140 at ds = 32; both training logs reach epoch 250. Appendix A records the attention epochs and other training settings.

6.4 Dense-observation background and other systems

Earlier Easy Attention experiments explored denser KS observations, ds = 1, 2, 4 and 8, corresponding to 64, 32, 16 and 8 measured positions. Their mean NRMSE curves in Appendix B start at low values and increase during autonomous rollout. Those experiments motivated the later sparse settings of four and two channels. They used a window-level split and the earlier snapshot-dependent NRMSE denominator, so their curve heights describe a different evaluation procedure.

The two supplementary systems extend the forecasting workflow to different dynamics and state representations. For Lorenz–96, the NRMSE comparison uses 20 recorded independent test trajectories; for two-dimensional Kolmogorov vorticity it uses 15. Table 4 gives first crossings computed from each individual test-trajectory curve. Time is expressed in each system's model-time units (saved steps × 0.01); the Kolmogorov Lyapunov setting remains uncalibrated.

System	Procedure	$VPT_{0.5}$ (model time)	VPT_1 (model time)
Lorenz–96	Self-attention	1.9520	2.6825
Lorenz–96	Easy Attention + feature gate	1.9930	2.8345
Lorenz–96	LSTM	1.0835	1.9750
2D Kolmogorov	Self-attention	0.5233	1.2980
2D Kolmogorov	Easy Attention + feature gate	0.4980	1.2347
2D Kolmogorov	LSTM	0.5253	1.3620

Table 4. Supplementary independent-test first-crossing means at NRMSE 0.5 and 1, in each system's model time. Lorenz–96 has 20 test trajectories and 3,000 forecast steps; two-dimensional Kolmogorov has 15 and 2,000. Both save at 0.01 intervals.

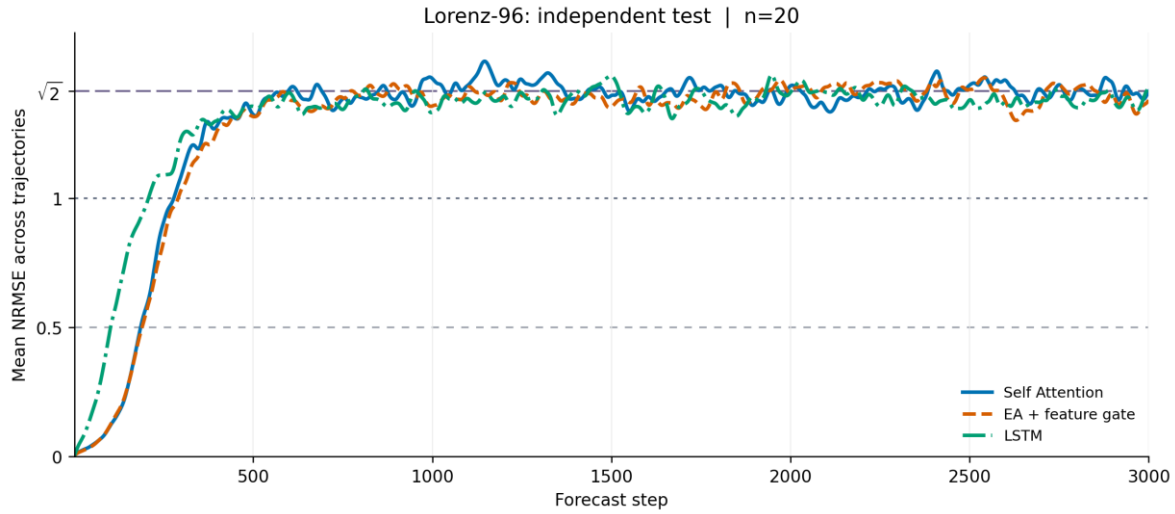


Figure 9. Lorenz–96 mean NRMSE over 20 recorded independent-test trajectories, with forecast lead in saved steps and guides at 0.5, 1 and √2. Curves use the recorded per-trajectory errors.

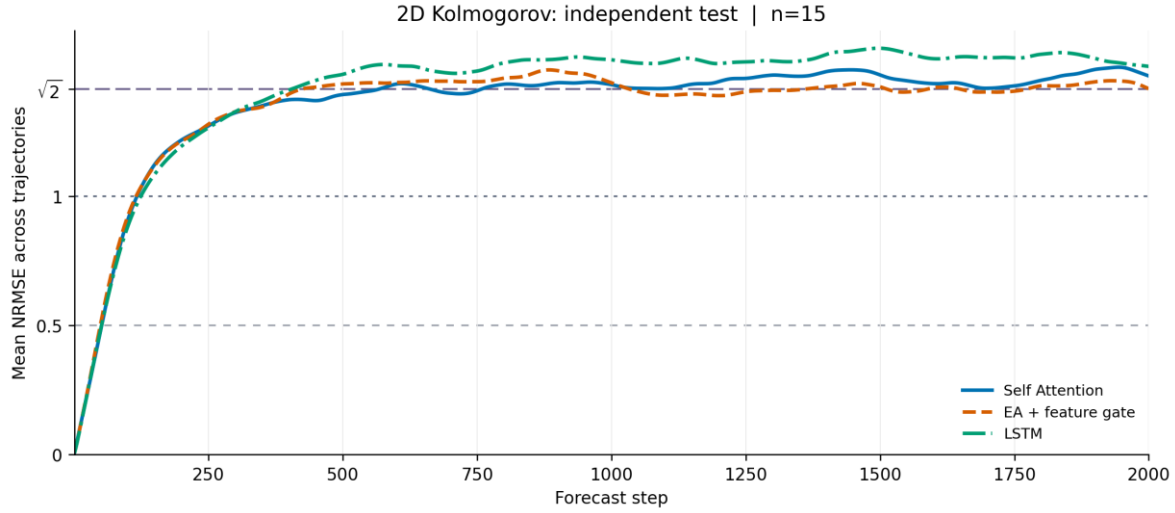


Figure 10. Two-dimensional Kolmogorov mean field NRMSE over 15 recorded independent-test trajectories, with forecast lead in saved steps and guides at 0.5, 1 and √2. Curves use the recorded per-trajectory errors.

For Lorenz–96, feature-gated attention has slightly longer mean VPT than self-attention at both tolerances; the LSTM is shorter. For the Kolmogorov field, the three early crossing times are close, and LSTM has the longest mean at the threshold of 1. These are system-specific observations from selected checkpoints. After the Table 4 first crossings, the mean curves in Figures 9 and 10 approach or fluctuate around √2.

7 Discussion

Under the recorded training and rollout procedures, the attention models maintain smaller pointwise errors for longer than LSTM at both sparse observation settings. Their relative order changes with ds and the chosen NRMSE tolerance. The thresholds 0.5 and 1 are operational standards in this study: VPT marks the first time an individual trajectory reaches one of those error levels. The later curve describes long-range error behavior separately from that first crossing.

The methodological changes also improve the meaning of the comparison. Splitting by trajectory keeps nearly duplicated windows in one data group. Feeding whole attention blocks uses all 100 directly supervised outputs at inference. Training-channel scales keep the NRMSE denominator fixed with lead time, especially important when only two spatial channels are observed. Closed-loop LSTM training

includes the model's own feedback for progressively longer intervals. A controlled ablation on one split and error definition would measure each change's contribution to forecasting skill.

The 20 held-out KS trajectories describe variation across forecast starts. The models differ in architecture, optimizer, batch size and feedback objective. At ds = 32, LSTM also has a longer warm-up and usually a shifted evaluation window. KS results use validation trajectories involved in checkpoint selection; the other systems have separately labeled test trajectories. Appendix A records the scope of the source-data and metric checks.

A longer-term question is whether a forecast can preserve useful statistics after losing its exact trajectory. The present plots cover NRMSE and one observed-state trace. Testing amplitude distributions, temporal correlations or energy spectra after VPT would answer a different question. That work would also benefit from aligned ds = 32 forecast starts and repeated training seeds.

8 Conclusions and Future Work

This project developed a traceable workflow for autonomous forecasting of observed chaotic states. The report reconstructed the initial KS pipeline, identified where overlapping-window validation and a changing snapshot-variance denominator obscured interpretation, and explained the revised trajectory split, 100-step attention blocks, fixed training-channel normalization and LSTM closed-loop horizon schedule. The selected forecasts were then evaluated from their individual unsmoothed NRMSE curves, with VPT at both 0.5 and 1 in reference KS Lyapunov time.

At ds = 16, feature-gated Easy Attention has the longest VPT at both reported tolerances. At ds = 32, it leads self-attention at the stricter 0.5 tolerance, while self-attention leads at the more permissive threshold of 1. The ds = 32 LSTM has a short initial VPT and fails as a long autonomous pointwise predictor despite its late mean NRMSE remaining below $\sqrt{2}$. All three KS procedures eventually lose exact long-horizon alignment. Lorenz–96 and two-dimensional Kolmogorov extend the analysis to a cyclic state and a vorticity field, with results reported on their own time scales.

The next KS experiment should reserve a separate test group and record its initial-condition seed. At ds = 32, the models should forecast the same future windows: LSTM can receive its preceding 256 states and the attention models their preceding 128, with the first predicted time aligned. Repeated training seeds would show the stability of the observed VPT differences. A controlled series could vary history length, attention block length, feature gate or recurrent horizon curriculum one at a time. Measured inference time would complement model-call counts. Post-VPT amplitude distributions, temporal correlations and spatial spectra could then test whether the forecasts retain useful statistics after phase agreement is lost.

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Appendix A Verified Experimental Parameters

A.1 Numerical data and evaluation

The KS configuration specifies a periodic domain of length 22, a 64-point numerical grid and saved states every $\Delta t = 0.1$ model-time unit. Spatial sampling retains every sixteenth or thirty-second point, giving four and two observed channels. The 100-trajectory record contains 100,000 saved time positions per trajectory. Training windows advance with stride one. Seed 42 assigns 80 complete trajectories to training and 20 to validation before any windows are cut. Channel means and population standard deviations are fitted on the training trajectories and then held fixed for inference and NRMSE evaluation. The seed of the numerical initial-condition draw was not retained with the run record.

Each selected model is forecast for 3,000 saved steps on the 20 held-out validation trajectories. The predictor chooses one deterministic start per trajectory, spreading starts across the available time range. For a 128-step warm-up the recorded starts range from 0 to 96,872; the 256-step ds = 32 LSTM warm-up ranges from 0 to 96,744. The trajectory IDs agree across the six runs, while the longer LSTM warm-up shifts most of its ds = 32 truth windows. NRMSE first computes the channel-normalized RMS error for each trajectory and forecast step, using the fixed training-channel standard deviations, then takes the mean across the 20 trajectories. The plotted Lyapunov-time conversion uses the literature reference $\lambda = 0.043$ [9], so one saved KS step corresponds to 0.0043 reference Lyapunov times. The exponent was adopted for the time axis; it was not estimated from these trajectories.

The archived KS trajectories reproduce every evaluated validation window to float32 precision; training-set channel statistics computed from them reproduce the six stored NRMSE arrays exactly. For Lorenz–96 and Kolmogorov flow, the local export retains per-trajectory errors for recomputing the plotted summaries and first crossings, while the original train and test trajectory files remain outside that export.

A.2 Model and training records

Table A1 identifies the saved model chosen for each figure and the trainable parameter count calculated from its implemented layers. Fixed sinusoidal position buffers are excluded. All four attention logs reach

epoch 600, and both LSTM logs reach epoch 250; the selected epochs in the table come from validation rollout checkpoint selection.

Model	ds	Channels	History	Selected epoch	Parameters
Self-attention	16	4	128	350	2,539,152
Feature-gated Easy Attention	16	4	128	310	2,734,992
LSTM	16	4	128	245	1,321,988
Self-attention	32	2	128	390	2,487,240
Feature-gated Easy Attention	32	2	128	480	2,683,080
LSTM	32	2	256	140	1,319,426

Table A1. KS model dimensions, selected epochs and trainable parameter counts.

Both attention models use three 256-wide encoder layers, eight heads, a 1,024-wide feed-forward network and dropout probability 0.001. A 128-step observed history directly produces a 100-step target block. During forecasting, the next block receives the newly predicted history. The feature-gated Easy Attention model additionally learns one $8 \times 128 \times 128$ temporal matrix per layer, initialized near the identity with Gaussian noise of standard deviation 0.01. Attention training uses uniformly weighted 100-step MSE, batch size 32, AdamW, initial learning rate 10^{-4}, weight decay 10^{-4} and a gradient-norm limit of 1. The learning rate follows a cosine schedule to 10^{-6}. Each epoch uses at most 500 training batches and 100 fixed validation batches; early-stopping patience is configured at 300 epochs.

The LSTM stacks three 256-unit recurrent layers and predicts one state per call while carrying hidden and cell states through feedback. Its training settings are batch size 64, Adam, zero weight decay and a gradient-norm limit of 0.5. During training, input dropout has probability 0.005 and zoneout retains the previous returned state with probability 0.02. Both effects are disabled for the plotted inference. Each epoch uses up to 100 training batch groups. The model remains in float32 and automatic mixed precision is disabled, as in the attention runs. The LSTM's 40-epoch early-stopping counter is active from epoch 101; the saved logs reach the 250-epoch maximum.

Epochs	Closed-loop horizon H	Learning rate
1–20	10 steps	5×10^{-4}
21–50	20 steps	3×10^{-4}
51–100	50 steps	2×10^{-4}
101–250	100 steps	Cosine: 10^{-4} to 2×10^{-5}

Table A2. LSTM closed-loop training curriculum in the completed ds = 16 and ds = 32 runs.

At each horizon change, the current-H loss measures a longer rollout and can rise even while optimization continues within a stage. The training logs also record a fixed 100-step validation loss. Validation rollouts over 1,200 steps are checked every ten attention epochs and every five LSTM epochs when selecting the saved forecasting checkpoint. The final 3,000-step KS curves show these selected models on held-out validation trajectories. A separate KS test would provide an additional check of generalization.

A.3 Supplementary system settings

System	Numerical setup	Evaluation
Lorenz–96	40 variables; forcing F = 8; RK4 step and saved interval 0.01	20 independent tests; 3,000 steps; selected epochs 50 / 25 / 115
2D Kolmogorov	64 × 64 vorticity; viscosity 0.005; drag 0.1; forcing wavenumber 4; integrator step 0.005; saved interval 0.01	15 independent tests; 2,000 steps; selected epochs 95 / 95 / 30

Table A3. Separate test systems. Selected epochs are ordered self-attention, feature-gated Easy Attention, LSTM.

Lorenz–96 has 100 trajectories in its train/validation pool, and Kolmogorov has 75. The Kolmogorov networks include a spatial encoder and decoder; its LSTM result is from a 30-epoch, 200-step fine-tune. For Lorenz–96, $\lambda = 1.68$ [2] is a literature reference adopted in the configuration. The Kolmogorov exponent is explicitly uncalibrated, so that system is described in forecast steps and physical model time. The three systems' reported horizons belong to distinct dynamical tasks and test sets.

Appendix B Earlier Exploratory KS Runs

The earlier Easy Attention experiments examined denser spatial observations at ds = 1, 2, 4 and 8. Each model used 64 observed time steps to supervise three future steps. The prediction script then fed back its first output at each forecast call. Windows of length 64 were cut with stride eight from 100 trajectories; adjacent input windows therefore shared 56 values. The validation split followed window extraction, and the evaluation trajectories overlapped those used in training. This stage revealed the need to separate trajectories before creating windows.

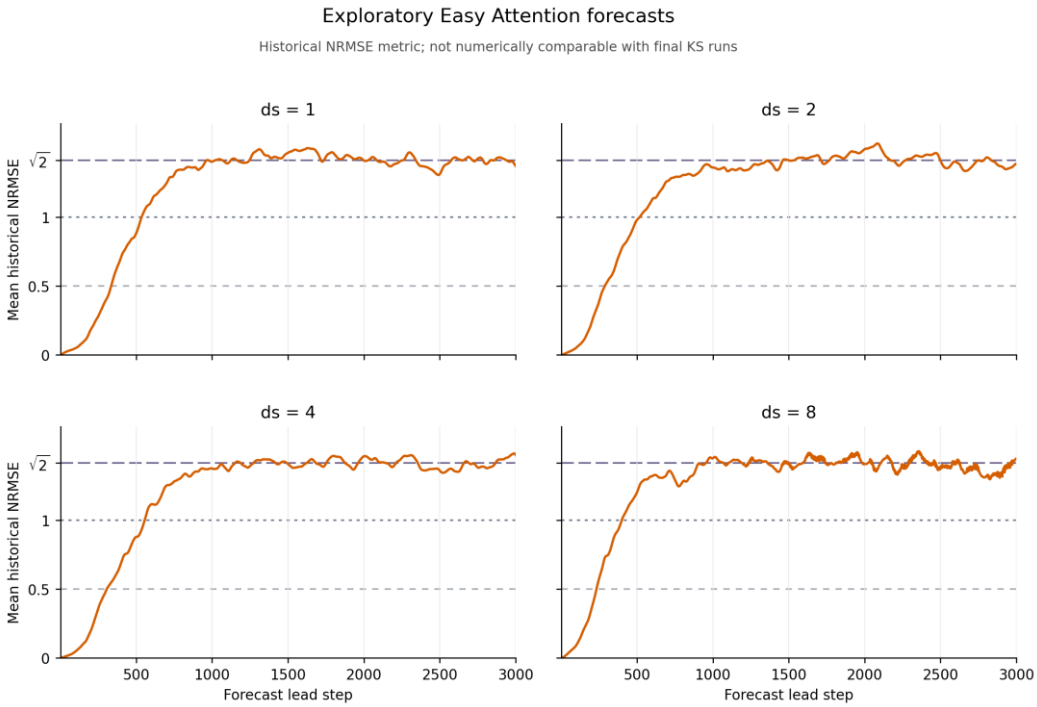


Figure B1. Earlier Easy Attention mean NRMSE at ds = 1, 2, 4 and 8. Horizontal lines mark 1 and √2. The calculation used the truth's spatial variance at each forecast step as its denominator.

The earlier denominator changed with the true spatial snapshot, and the four spatial-sampling data files have no recorded pairing of initial conditions. These curves document the exploratory stage; the final comparison uses a fixed training-channel scale and held-out trajectories. Its VPT analysis starts from each final trajectory's own first threshold crossing, which a group-mean curve cannot supply.